



Assignment Brief

Higher National Diploma in Computing

Unit 5: Security

Student Name/ID Number	
Unit Number and Title	Unit 5: Security
Academic Year	2026
Unit Tutor	Mr. Aung Bo Bo Kyaw
Assignment Title	Assignment 1 of 1: Designing a Defense-in-Depth Security Posture and Access Control Architecture for MetroMart & StarLight Enterprise Group
Issue Date	21 th May 2026
Submission Date	19 th June 2026
Submission Format	
Your report must be formatted as follow: • Font Name: Verdana • Font Size: 12 pt • Margin: 0.75 inches (round) • Line Spacing: 1.5 pt. • Alignment: Justify • Left hand side of header: Batch Number	

- Right and side of header: Your name
- Centre of footer: Page Number
- Use the Harvard referencing system
- Complete the title page and sign the statement of authenticity
- Your assignment must be submitted in pdf format

Unit Learning Outcomes

The purpose of this assignment is to:

L01 Access risks to IT security

L02 Describe IT security solutions

L03 Review mechanisms to control organisational IT security

L04 Manage organisational security

Transferable skills and competencies developed

On successful completion of this unit, students will be able knowledge, proficiency, and an understanding of security fundamentals, enabling you to analyse vulnerabilities and implement effective security measures.

Vocational scenario

You are employed as a Junior IT Security Team at **TechSecure & BIM Cyber Solutions Ltd**, a leading cybersecurity and network defence provider in Myanmar. Your company offers a comprehensive suite of services, including information security consulting, cyber defence, vulnerability testing, and network infrastructure protection.

Recently, a major conglomerate, **MetroMart & StarLight Enterprise Group**, contacted your firm. Due to the rapid digitalization of their business operations (including online booking systems, digital payment gateways, delivery operations, guest Wi-Fi, and customer loyalty databases), they are facing an increasing volume of cybersecurity challenges and unauthorized access attempts within their internal network.

The client's primary headquarters is a multi-story facility designed with the following physical layout:

- 3rd Floor: Office & Management Department
- 2nd Floor: IT, Finance, and Admin Department
- 1st Floor: Market / Retail Zone
- Lobby: Restaurant, Guest Area & Public Wi-Fi Zone

As the assigned Cyber Defence Team, your primary objectives are to perform a comprehensive risk assessment, draft organizational security policies, and develop a functional network prototype that demonstrates the strategic layout and implementation of robust **Access Control List (ACL)** rules to secure the corporate infrastructure.

Assignment activity and guidance

This assignment requires the collective efforts of a student group to produce and submit three core components: a **Group Presentation**, a **Network Prototyping Lab File**, and an **Overall Technical Report**. The specific guidelines and structural requirements for each component are outlined below:

Task 1: Group Presentation & Recording

The team is required to design a professional presentation slide deck containing **20 to 30 slides** covering the business and risk management aspects of the project. The structure must include:

- **Security Risk Analysis:** A detailed discussion on the types of security risks applicable to the enterprise and a review of existing organizational security procedures.
- **Risk Management Framework:** An overview of how IT security risks can be assessed and treated, proposing a structured methodology aligned with **ISO 31000** principles or relevant risk management standards.
- **Operational Monitoring:** Discussion slides outlining at least three distinct benefits, with clear justifications, for implementing automated **Network Monitoring Systems (NMS)**.

Task 2: Network Prototyping Lab File

The student group must build a functional network prototype that maps out the enterprise's multi-story facility layout. This simulation file must explicitly demonstrate:

- **Network Infrastructure Segregation:** A clear logical layout separating the facility's departments (Management, IT, Finance, Retail, and the Public Lobby/Guest Area) into distinct subnets to satisfy internal data protection guidelines.
- **Access Control List (ACL) Rules:** Implementation of robust standard or extended ACL rules to enforce strict traffic filtering based on the organization's security requirements, highlighting the security impact if rules are misaligned.
- **Configuration Failures Analysis:** A section or configured scenario demonstrating/explaining the severe security impacts on the infrastructure if firewall policies and third-party VPNs were configured incorrectly.

- **Security Enhancements:** Practical examples showing how network defense is improved through the strategic deployment of a **DMZ, Static IP, and Network Address Translation (NAT)**.

Task 3: Overall Technical Report

The final deliverable is a comprehensive, cohesive technical report compiled by the group that synthesizes the engineering choices, policy designs, and risk evaluations. It must feature:

- **Security Policy Architecture:** A review of organizational risk assessment procedures and a tailored corporate **Security Policy**, accompanied by operational core components of a **Disaster Recovery (DR) Plan** and a **Business Continuity Plan (BCP)** with clear justifications.
- **Defense-in-Depth Evaluation:** A rigorous evaluation of at least three physical and three virtual security measures implemented within the prototype to ensure the overall integrity and suitability of tools to meet business needs.
- **Audit and Compliance:** An analysis of the possible impacts resulting from a formal IT security audit, along with a discussion on the strategic roles that organizational stakeholders play during these security audits.

Evidences

You should submit technical report in **pdf file, simulation lab file and Presentation slide.**

Note to Students:

- Check carefully the submission date and the instructions given with the assignments
- Late assignments will not be accepted, except for valid circumstances with supporting evidences
- Ensure that you give yourself enough time to complete the assignment by the due date.
- Do not leave things such as printing to the late minute-excuses of this nature will not be accepted for failure to hand-in the work on time.
- You must take responsibility for managing your own time effectively.
- If you are unable to hand in your assignment on time and have valid reasons such as illness, you may apply (in writing) for an extension.
- Take great care that if you use other people's work or ideas in your assignment, you properly reference them in your text and any bibliography.

Note: Plagiarism is on offence. Plagiarized work will be differed

Interim Review Session (Formative Assessment)

- 3rd June 2026 (Wednesday)
- 4th June 2026 (Thursday)

Submit the work to moodle.gusto-education.com by

- 11:55 PM on 19th June 2026 (Friday)

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
L01 Assess risks to IT security		D1 Evaluate a range of physical and virtual security measures that can be employed to ensure the integrity of organisational IT security.
P1 Discuss types of security risks to organisations. P2 Assess organisational security procedures.	M1 Analyse the benefits of implementing network monitoring systems with supporting reasons.	
L02 Describe IT security solutions		
P3 Discuss the potential impact to IT security of incorrect configuration of firewall policies and third- party VPNs. P4 Discuss, using an example for each, how implementing a DMZ, static IP and NAT in a network can improve network security.	M2 Propose a method to assess and treat IT security risks.	
L03 Review mechanisms to control organisational IT security		D2 Recommend how IT security can be aligned with an organisational policy, detailing the security impact of any misalignment.
P5 Review risk assessment procedures in an organisation. P6 Explain data protection processes and regulations as applicable to an organisation.	M3 Summarise an appropriate risk management approach or ISO standard and its application in IT security. M4 Analyse possible impacts to organisational security resulting from an IT security audit.	
L04 Manage organisational security		
P7 Design a suitable security policy for an organisation, including the main components of an organisational disaster recovery plan. P8 Discuss the roles of stakeholders in the organisation in implementing security audits.	M5 Justify the security plan developed giving reasons for the elements selected.	D3 Evaluate the suitability of the tools used in the organisational policy to meet business needs.